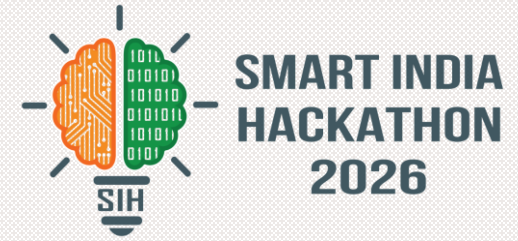


SMART INDIA HACKATHON 2026



TITLE PAGE

- **Problem Statement ID:** 26057
- **Problem Statement Title:** AI-Powered Automated Underwater Marine Debris and Anomaly Detection System using Side-Scan Sonar Imagery
- **Theme:** Disaster Management
- **PS Category:** Software
- **Team Name:** Team Unstable



THE QUESTION ISN'T JUST
“WHAT’S DOWN THERE?”
IT’S **“WHAT MATTERS —
AND WHERE IS IT?”**



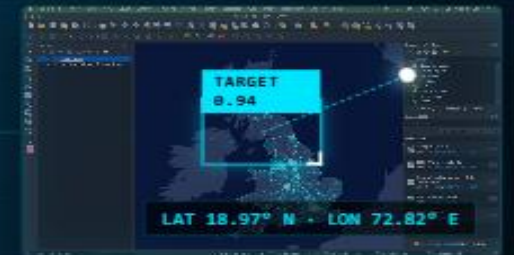
ANVESHAN

अन्वेषण

DETECT → VALIDATE → LOCATE

UNDERSTOOD

TRUST



ACTIONABLE
LOCATE

That is where Anveshan begins.

SONAR INPUT → PREPROCESSING → ULTRALYTICS YOLOv8





FEASIBILITY AND VIABILITY



SMART INDIA
HACKATHON
2026

ANVESHAN अन्वेषण

SIH 2026

FEASIBILITY & VIABILITY

TECHNICALLY ACHIEVABLE | EXISTING AI STACK | SCALABLE PIPELINE | FIELD-READY POTENTIAL

Side-Scan Sonar → AI Detection → Geospatial Intelligence

TECHNICAL FEASIBILITY



01 EXISTING SONAR DATA

Real SSS imagery already available via public datasets + synthetic augmentation.



02 PROVEN COMPUTER VISION

Ultralytics YOLOv8: mature object-detection architecture fine-tuned for sonar targets.



03 OPEN-SOURCE STACK

Python + OpenCV for robust preprocessing, enhancement, and automated inference.



04 AFFORDABLE COMPUTE

Local GPU training; prototype trained effectively on consumer-grade hardware.



05 DEPLOYABLE OUTPUT

Streamlit / Folium + JSON/CSV outputs for immediate geographic reporting.

● TECHNICALLY FEASIBLE



KEY CHALLENGES / RISKS



01 SONAR NOISE

Speckle, acoustic clutter, and variable contrast native to underwater environments.



02 FALSE POSITIVES

Natural seabed structures (rocks, ridges) may visually resemble artificial debris.



03 DOMAIN SHIFT

Inconsistent performance across different sensors, depths, and seabed conditions.



04 LIMITED LABELED DATA

Rare and specific debris classes are significantly underrepresented in training sets.



05 GEOLOCATION ACCURACY

Pixel-to-coordinate mapping strictly depends on the quality of survey metadata.

MITIGATION STRATEGIES



01 ROBUST PREPROCESSING

Automated despeckling, contrast enhancement, and nadir-gap masking applied early.



02 CONFIDENCE FUSION

Filtering by confidence score + shape regularity + acoustic-shadow consistency.



03 DATA AUGMENTATION

Injecting synthetic debris and variable acoustic noise into underrepresented sets.



04 MODEL CALIBRATION

Iterative threshold tuning and validation testing across diverse SSS condition silos.



05 GEOREFERENCING VALIDATION

Strict mathematical calibration for pixel → latitude/longitude transformation.

05 | IMPACT & BENEFITS

ANVESHAN
/अन्वेषण

Turning hidden sonar data into actionable marine intelligence



FROM SEARCHING THROUGH SONAR DATA → TO INTELLIGENTLY FINDING WHAT MATTERS



SEE THE UNSEEN. LOCATE IT. ACT ON IT.

RESEARCH AND REFERENCES

- <https://huggingface.co/datasets/PINGEcosystem/sss-crab-pot-detection-ds>
- <https://www.kaggle.com/datasets/enochkwatehdongbo/seabedobjects-klsg-dataset>
- <https://doi.org/10.5281/zenodo.7922705>